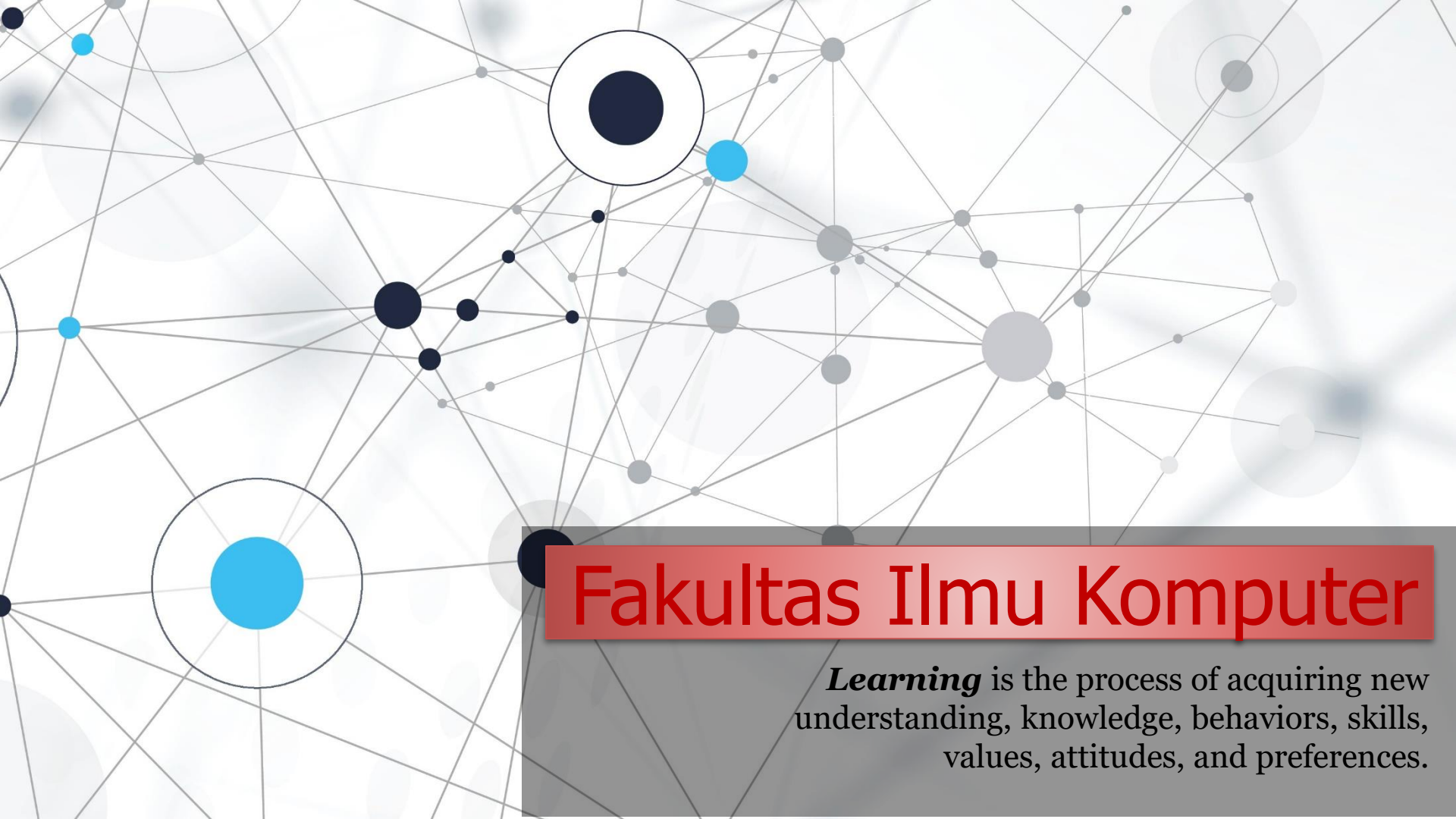


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

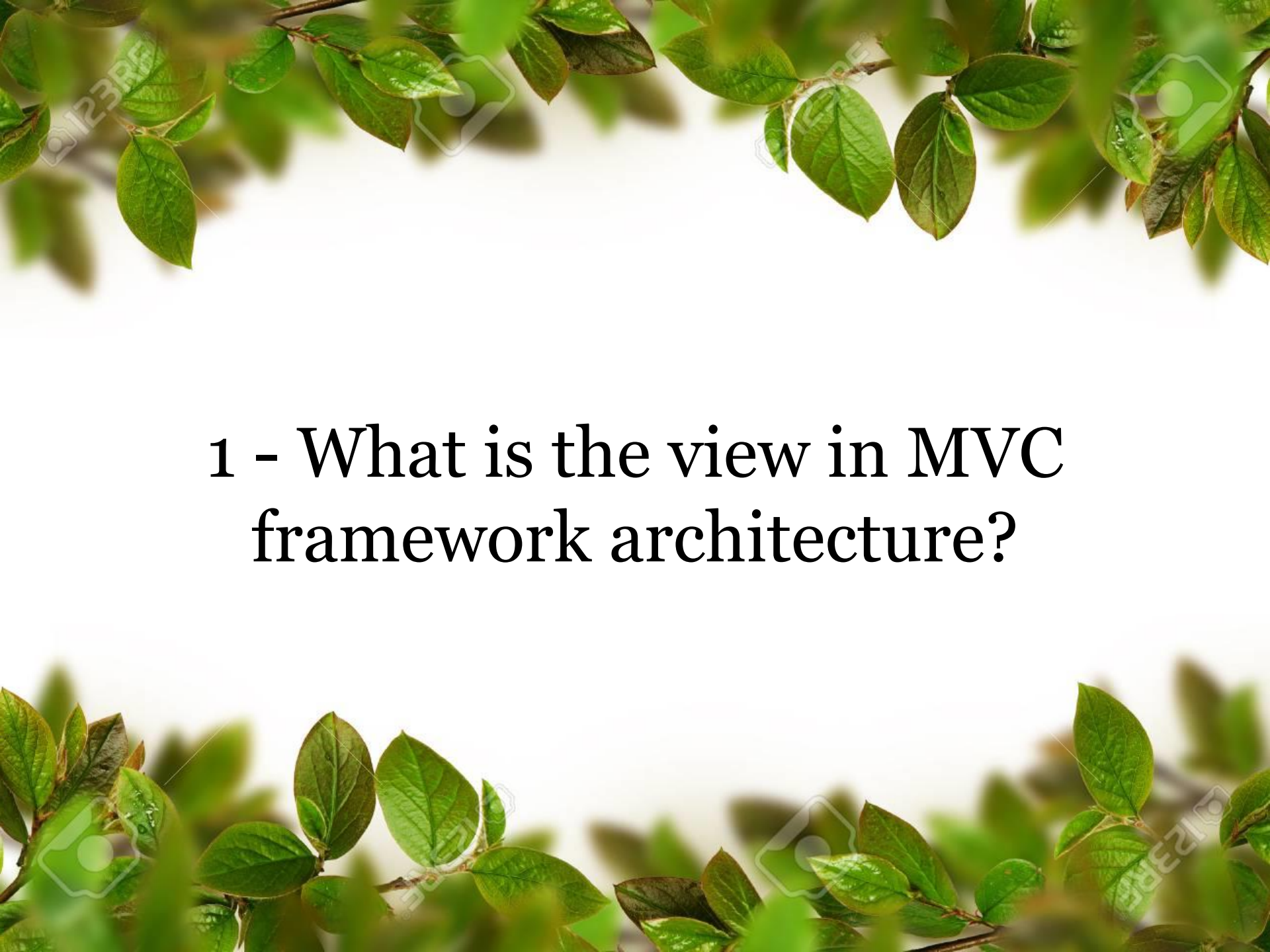
Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Framework #5 View



Welcome to the Framework Programming *course*



1 - What is the view in MVC
framework architecture?

Three Important MVC Components (*layers*)

Component #1 → Model :

- ✓ The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and send back to the database or use it to render the same data.
- ✓ It responds to the request from the views and also responds to instructions from the controller to update itself. It is also the lowest level of the pattern which is responsible for maintaining data.

Component #2 → View :

- ✓ The View is that part of the application that represents the presentation of data.
- ✓ Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.
- ✓ The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc.

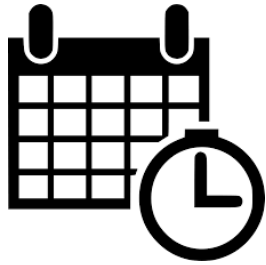
Component #3 → Controller :

- ✓ The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.
- ✓ A Controller send's commands to the model to update its state (*E.g., Saving a specific document*). The controller also sends commands to its associated view to change the view's presentation (*For example scrolling a particular document*).

The paper was published under the title:

Burbeck, Steve. "Applications programming in smalltalk-80 (tm): How to use model-view-controller (mvc)." *Smalltalk-80 v2 5* (1992): 1-11.

Three Important MVC Components (*layers*)



Model: It includes all the data and its related logic.



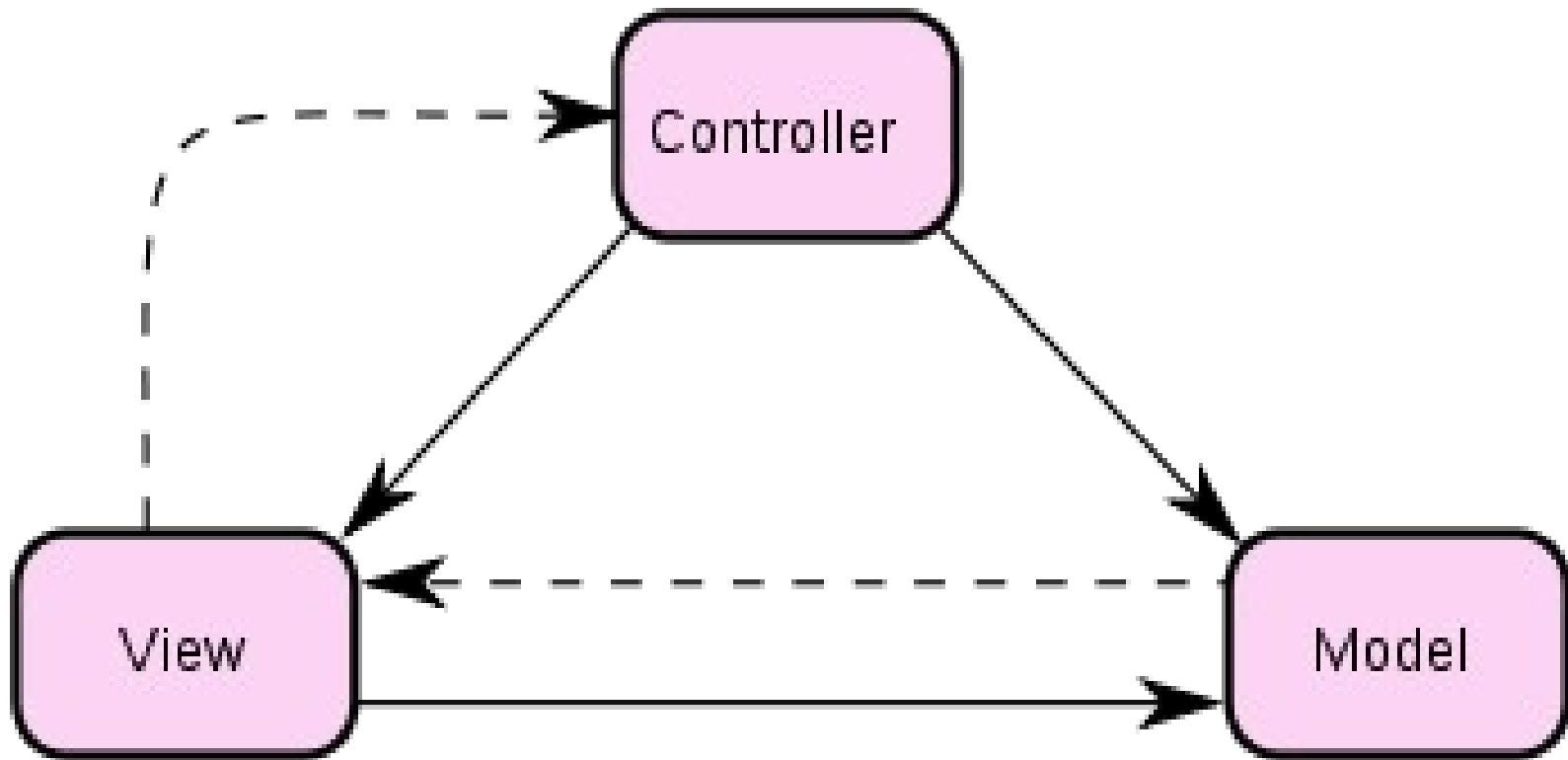
View: Present data to the user or handles user interaction.



Controller: An interface between Model and View components.

Three Important MVC Components (*layers*)

logic program (*controller*)



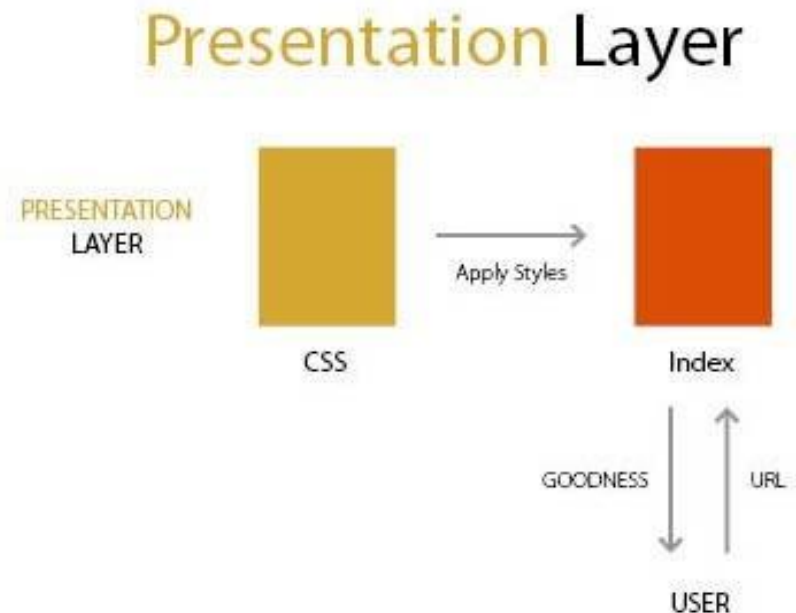
Tampilan (*view*) => UI

Manage the data

Component #2 $\Rightarrow$ View (Presentation)



Responsible to display the data provided by the model in a specific format.



A view is responsible for UI (*user interface*). View displays the data coming from the model.

Component #2 $\Rightarrow$ View (Presentation)



View in MVC is a user interface. A view is used to display data to the user using the model class object and also enables users to modify them.

Types of Views

1) Normal view

It's the same as web page. To create this view we use the view page template, (*MVC to view page template in MVC application*).

2) Partial view

It's the same as user control. Partial views are similar to the user controller.

3) Layout View

It's the same as a master page. To create a master page view we use a template.

Read Source:

<https://tekslate.com/many-types-view-mvc>

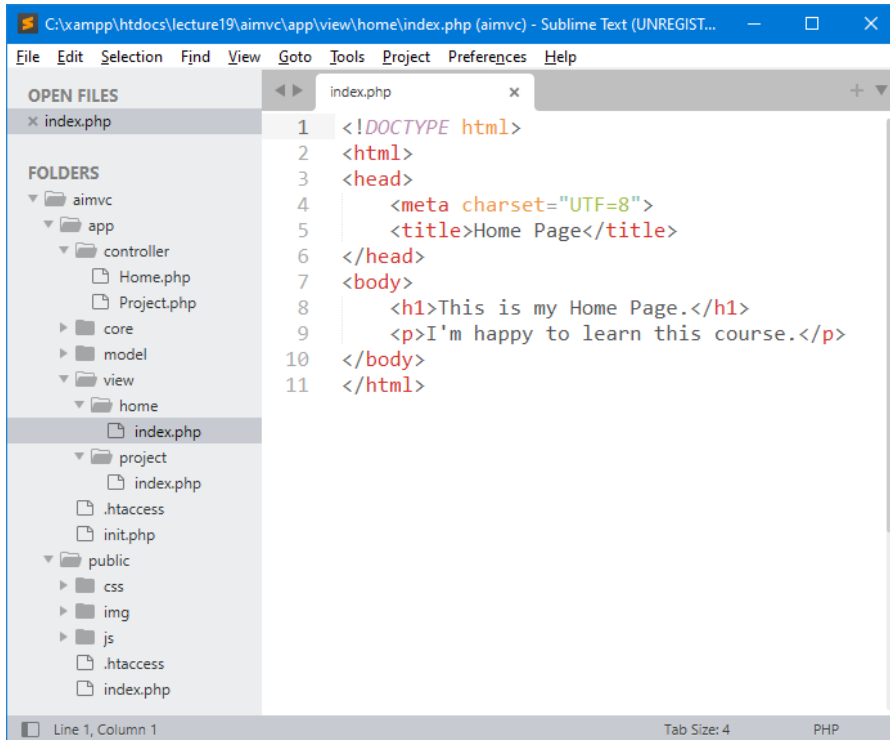
A decorative border of green leaves and branches at the top of the page.

2 – Basic ***display*** the page from
“*home & project*” view.

A decorative border of green leaves and branches at the bottom of the page.

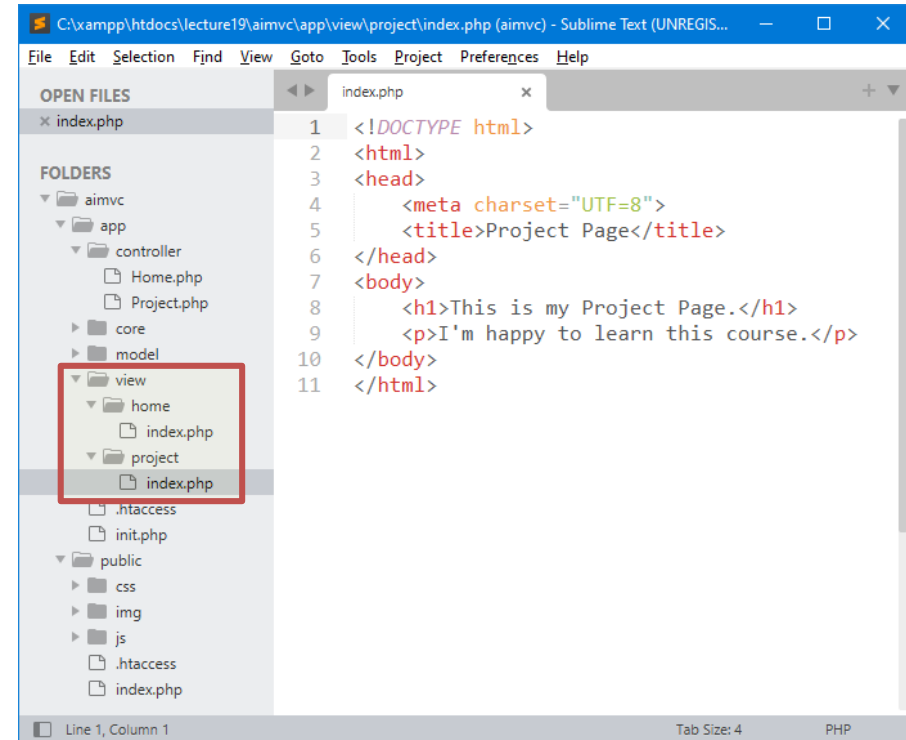
1 - Create Subdirectory in View

/view/home/index.php



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Home Page</title>
6 </head>
7 <body>
8   <h1>This is my Home Page.</h1>
9   <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

/view/project/index.php



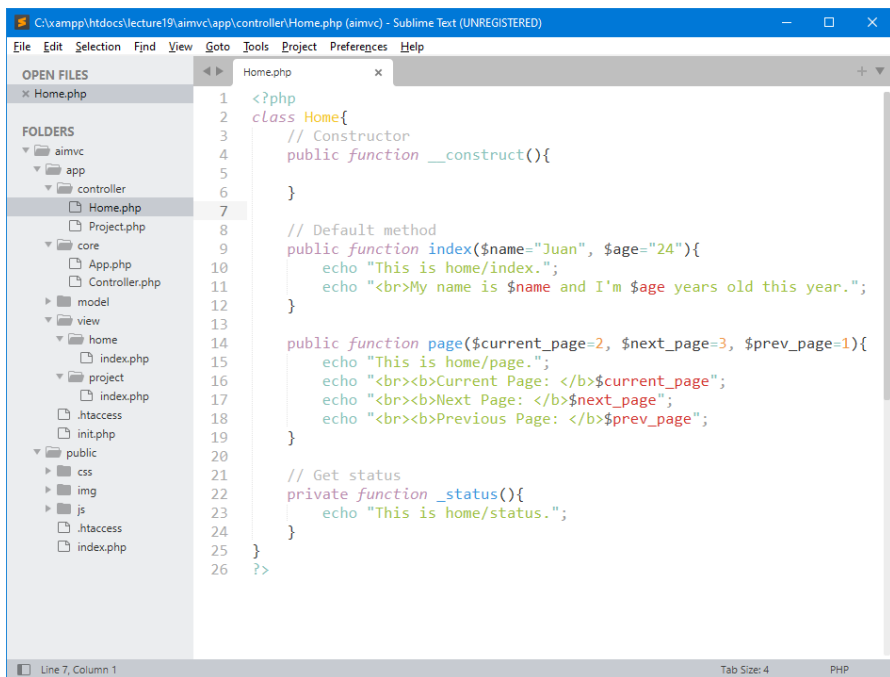
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Project Page</title>
6 </head>
7 <body>
8   <h1>This is my Project Page.</h1>
9   <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

- 1) The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- 2) The **<html>** element is the root element of an HTML page
- 3) The **<head>** element contains meta information about the HTML page
- 4) The **<title>** element specifies a title for the HTML page (*which is shown in the browser's title bar or in the page's tab*)
- 5) The **<body>** element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- 6) The **<h1>** element defines a large heading
- 7) The **<p>** element defines a paragraph

2 – Display Page from View

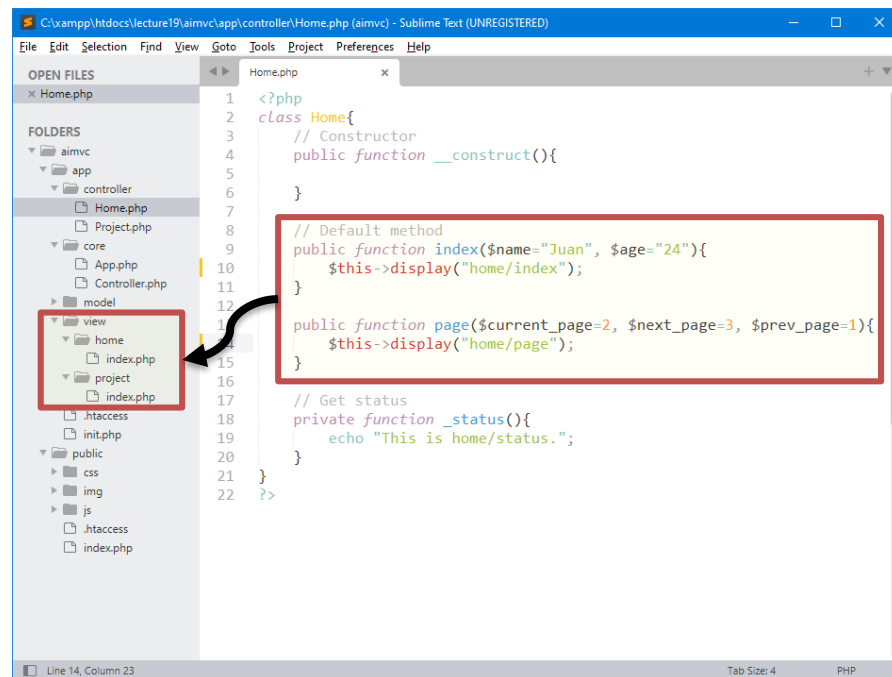
We handle this action from Controller. See, before and after changing code.

Code before change



```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page($current_page=2, $next_page=3, $prev_page=1){
15         echo "This is home/page.";
16         echo "<br><b>Current Page: </b>$current_page";
17         echo "<br><b>Next Page: </b>$next_page";
18         echo "<br><b>Previous Page: </b>$prev_page";
19     }
20
21     // Get status
22     private function _status(){
23         echo "This is home/status.";
24     }
25 }
26 ?>
```

Code after change



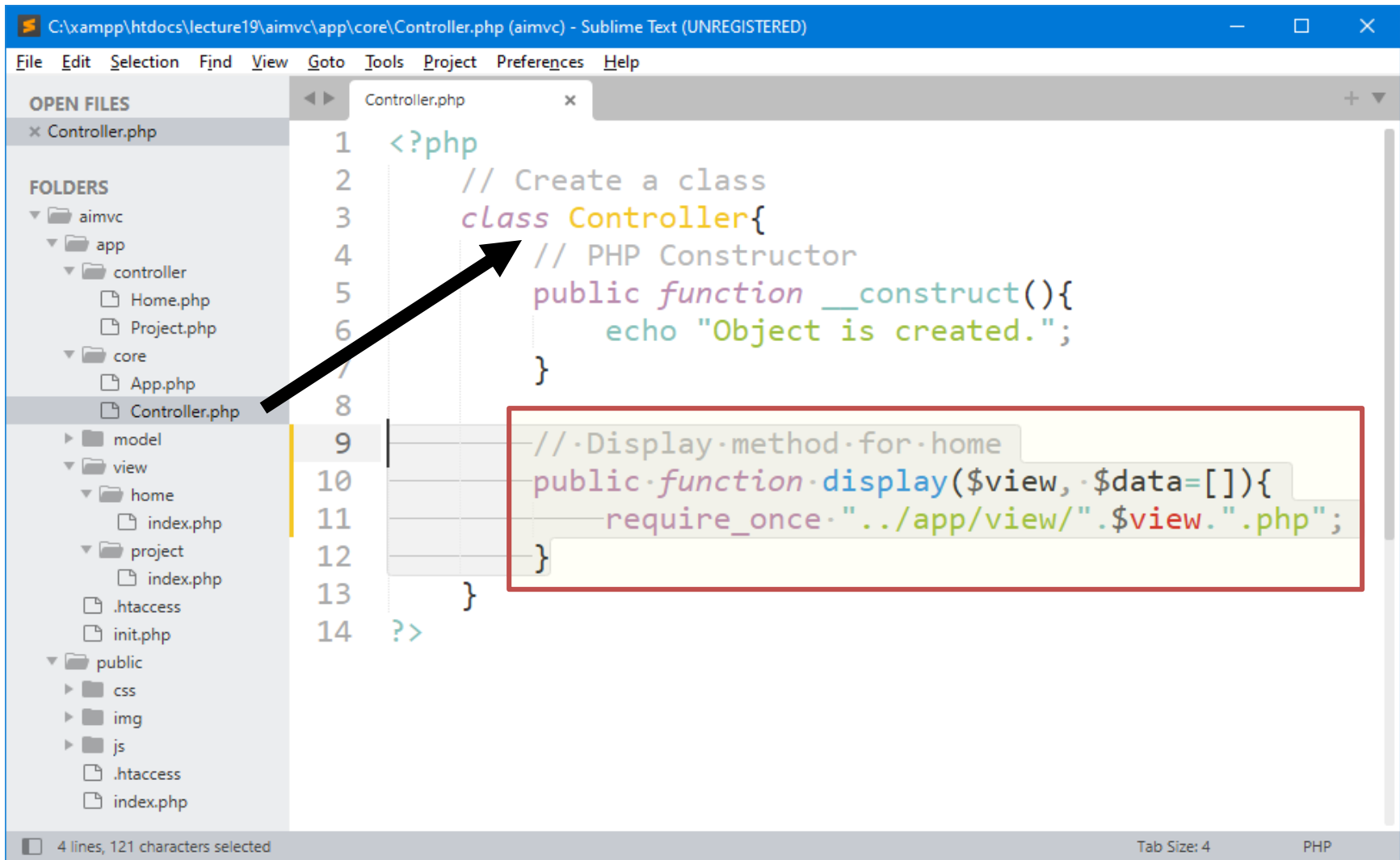
```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         $this->display("home/index");
11     }
12
13     public function page($current_page=2, $next_page=3, $prev_page=1){
14         $this->display("home/page");
15     }
16
17     // Get status
18     private function _status(){
19         echo "This is home/status.";
20     }
21 }
22 ?>
```

\$this->view("home/index");

Note, please see the selection file in left side tree menu. This code means you want to access ***index.php*** in home subdirectory which is located in view directory.

3 – Create Method *display()* in Core Controller

In variable “*view*” may contains value like “*home/index*”. This function will require file from ***view directory*** to get specific page and display it, like ***home*** page.



```
C:\xampp\htdocs\lecture19\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
Controller.php
x Controller.php
FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
      App.php
      Controller.php
    model
    view
      home
        index.php
      project
        index.php
      .htaccess
      init.php
    public
      css
      img
      js
      .htaccess
      index.php
1 <?php
2 // Create a class
3 class Controller{
4 // PHP Constructor
5 public function __construct(){
6     echo "Object is created.";
7 }
8
9 // Display method for home
10 public function display($view, $data=[]){
11     require_once "../app/view/".$view.".php";
12 }
13 }
14 ?>
```

4 lines, 121 characters selected

Tab Size: 4

PHP

4 – Inherit *display()* Function

Last step in this stage is to inherit this ***display()* function** to the controller class. The ***extends keyword*** is used to derive a class from another class. This is called inheritance.

```
C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Home.php

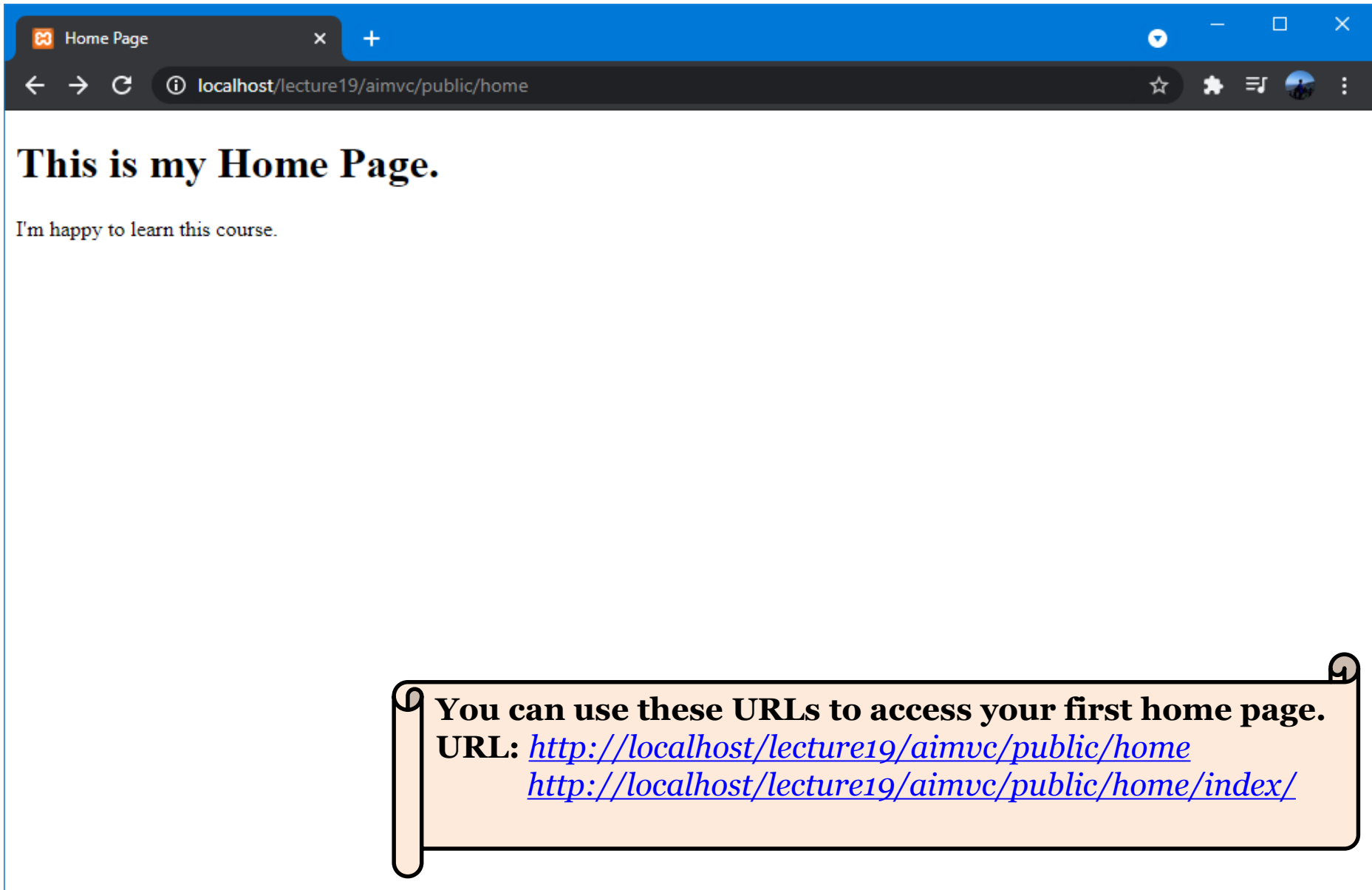
FOLDERS
  aimvc
    app
      controller
        Home.php
        Project.php
      core
        App.php
        Controller.php
      model
      view
        home
          index.php
        project
          index.php
      .htaccess
      init.php
      public
        css
        img
        js
        .htaccess
        index.php

1 <?php
2 class Home extends Controller{
3 // Constructor
4 public function __construct(){
6 }
7
8 // Default method
9 public function index($name="Juan", $age="24"){
10     $this->display("home/index");
11 }
12
13 public function page($current_page=2, $next_page=3, $prev_page=1){
14     $this->display("home/page");
15 }
16
17 // Get status
18 private function _status(){
19     echo "This is home/status.";
20 }
21 }
22 ?>
```

This controller class in core directory inherits all function to home class in controller directory.

30 characters selected Tab Size: 4 PHP

5 - Testing – View Home Page




```
<?php
class Home extends Controller{
    // Constructor
    public function __construct(){

    }

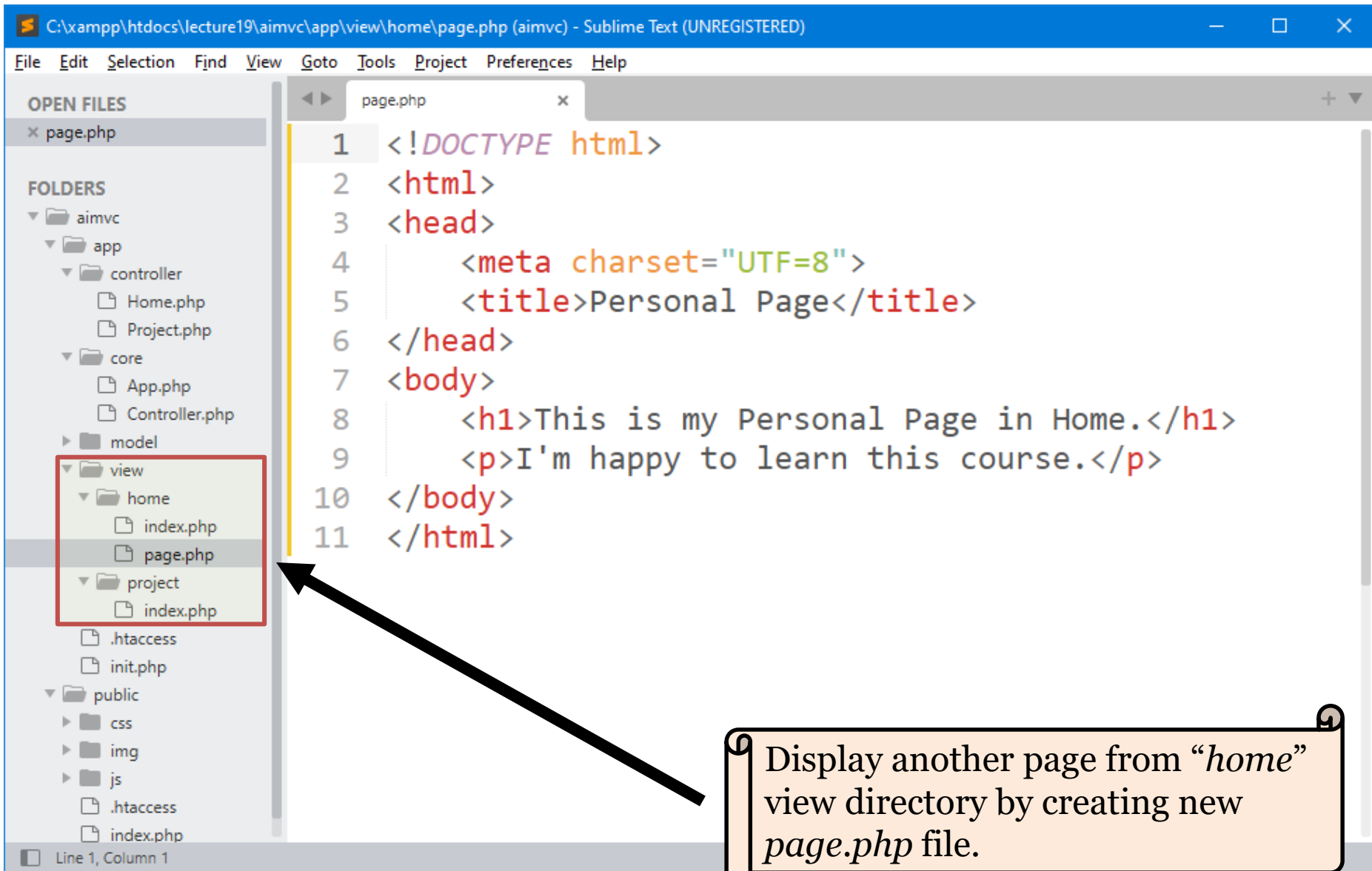
    // Default method
    public function index($name="Juan", $age="24"){
        $this->display("home/index");
    }

    public function page($current_page=2, $next_page=3, $prev_page=1){
        $this->display("home/page");
    }

    // Get status
    private function _status(){
        echo "This is home/status.";
    }
}
```

How to access this method *page()* in the home class? It will display an error if you force to access this method from the URL.

6 – Create *page.php* in *view/home* Directory



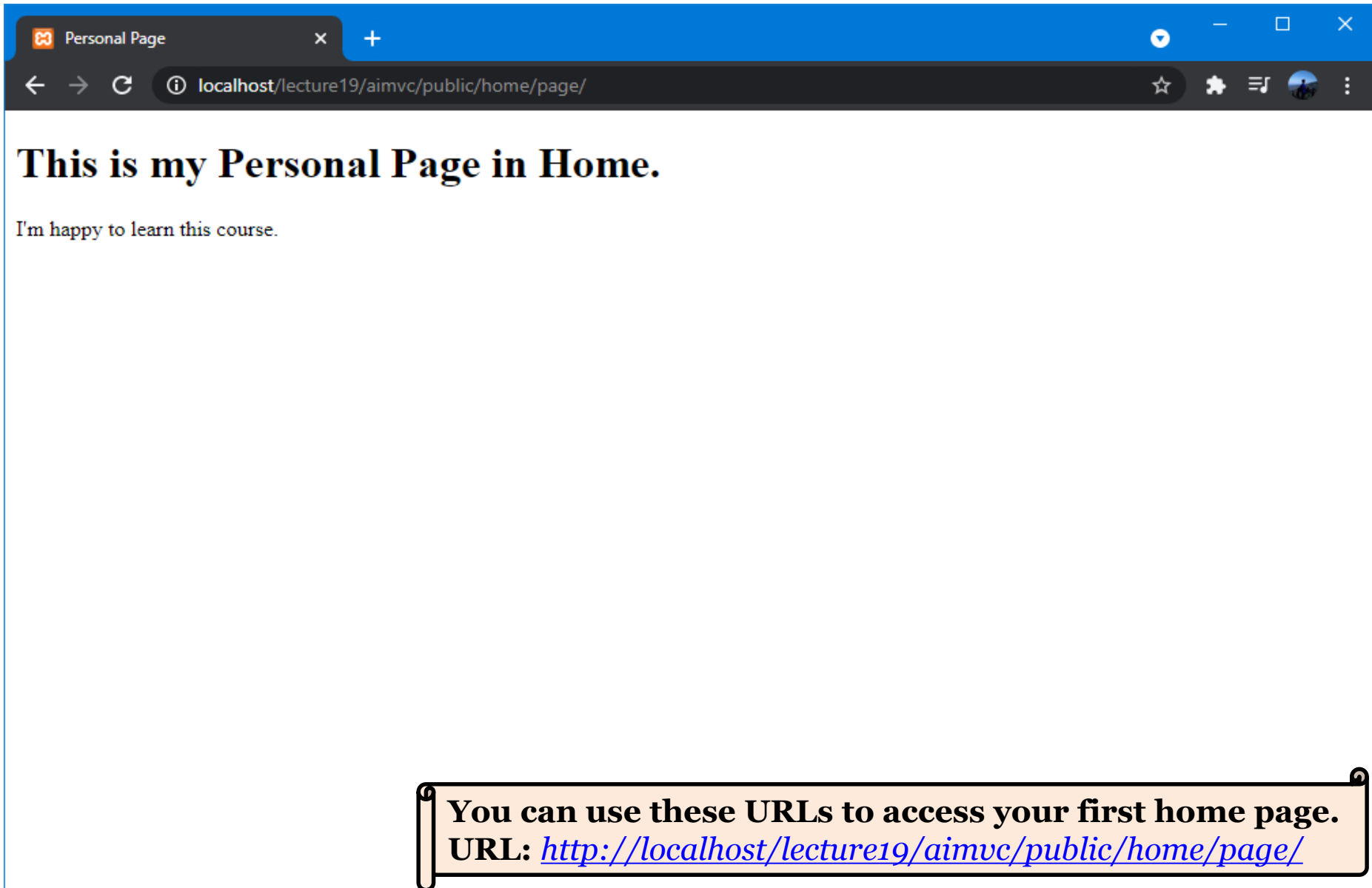
The screenshot shows the Sublime Text editor interface. The title bar indicates the file path: `C:\xampp\htdocs\lecture19\aimvc\app\view\home\page.php (aimvc) - Sublime Text (UNREGISTERED)`. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The left sidebar displays the 'FOLDERS' tree, where the 'view' directory is expanded, and 'home' is selected, showing 'index.php' and 'page.php'. A red box highlights the 'view' directory, and a black arrow points from a callout box to the 'page.php' file. The main editor area shows the content of 'page.php' with line numbers 1 through 11. The code is as follows:

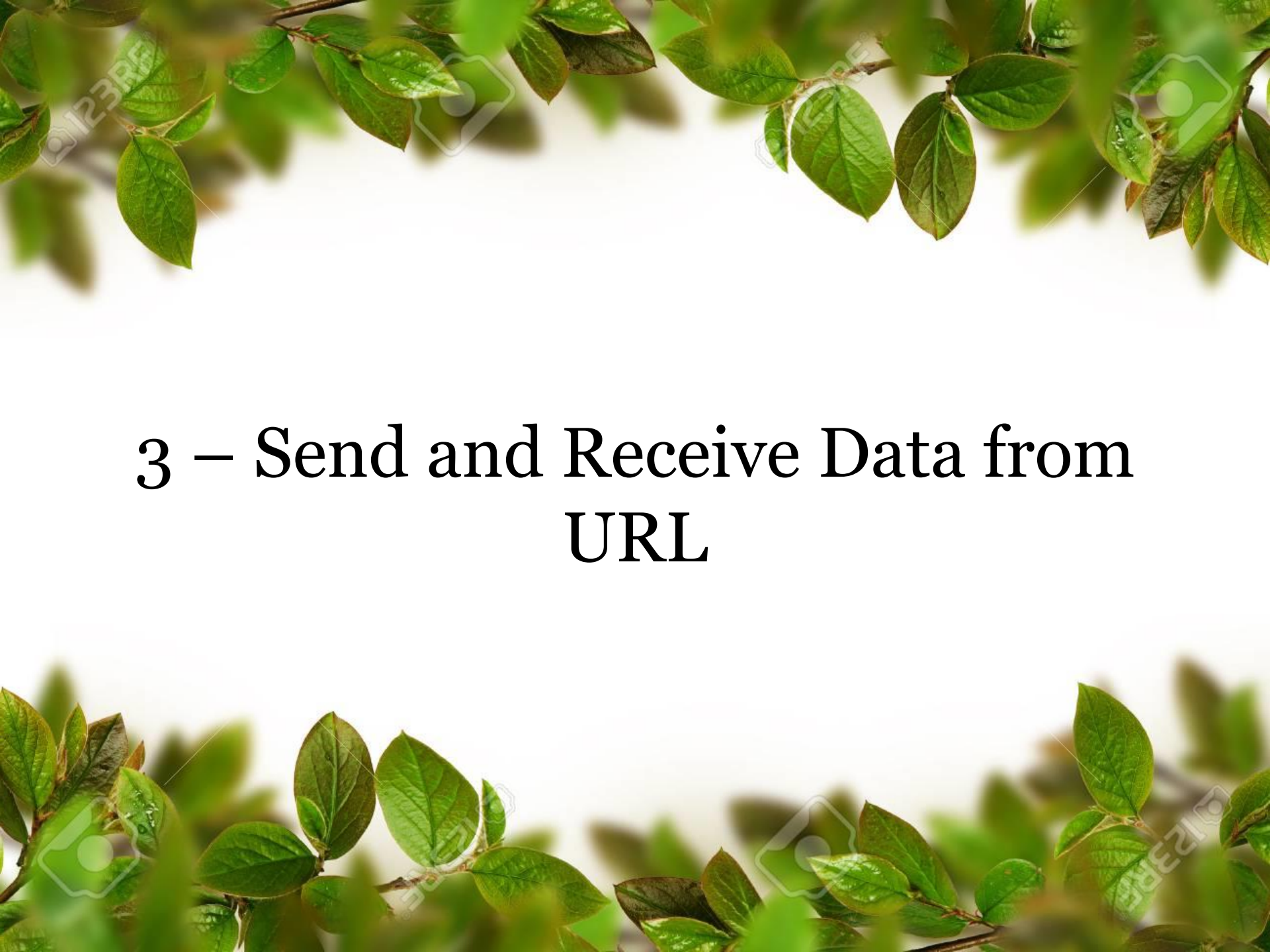
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9     <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

Line 1, Column 1

Display another page from “home” view directory by creating new *page.php* file.

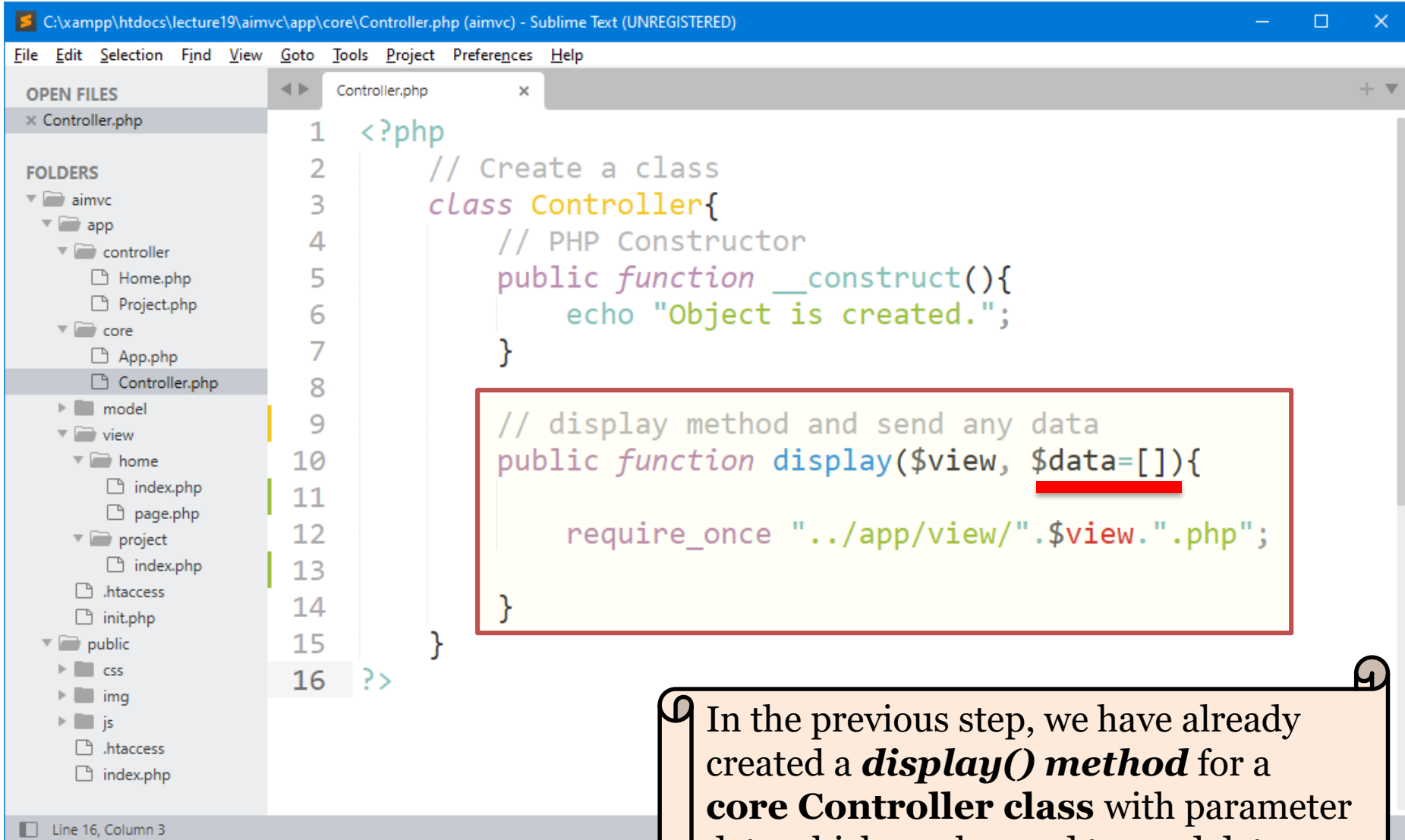
5 - Testing ⇒ View Personal Page





3 – Send and Receive Data from URL

1 – Check *display()* Method with Parameter Data



C:\xampp\htdocs\lecture19\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Controller.php

FOLDERS
aimvc
app
controller
Home.php
Project.php
core
App.php
Controller.php
model
view
home
index.php
page.php
project
index.php
.htaccess
init.php
public
css
img
js
.htaccess
index.php

```
1 <?php
2 // Create a class
3 class Controller{
4 // PHP Constructor
5 public function __construct(){
6     echo "Object is created.";
7 }
8
9 // display method and send any data
10 public function display($view, $data=[]){
11
12     require_once "../app/view/" . $view . ".php";
13
14 }
15 }
16 ?>
```

In the previous step, we have already created a ***display()* method** for a **core Controller class** with parameter data which can be used to send data.

Line 16, Column 3

2 - Assign an Associative Arrays to Method

The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the following PHP code:

```
1 <?php
2 class Home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         // Display page and send data
14         $this->display("home/index", $arr_data);
15     }
16
17     public function page($current_page=2, $next_page=3, $prev_page=1){
18         // Associative Arrays (arrays with keys)
19         $arr_data['current'] = $current_page;
20         $arr_data['next'] = $next_page;
21         $arr_data['previous'] = $prev_page;
22         // Display page and send data
23         $this->display("home/page", $arr_data);
24     }
25
26     // Get status
27     private function _status(){
28         echo "This is home/status.";
29     }
30 }
31 ?>
```

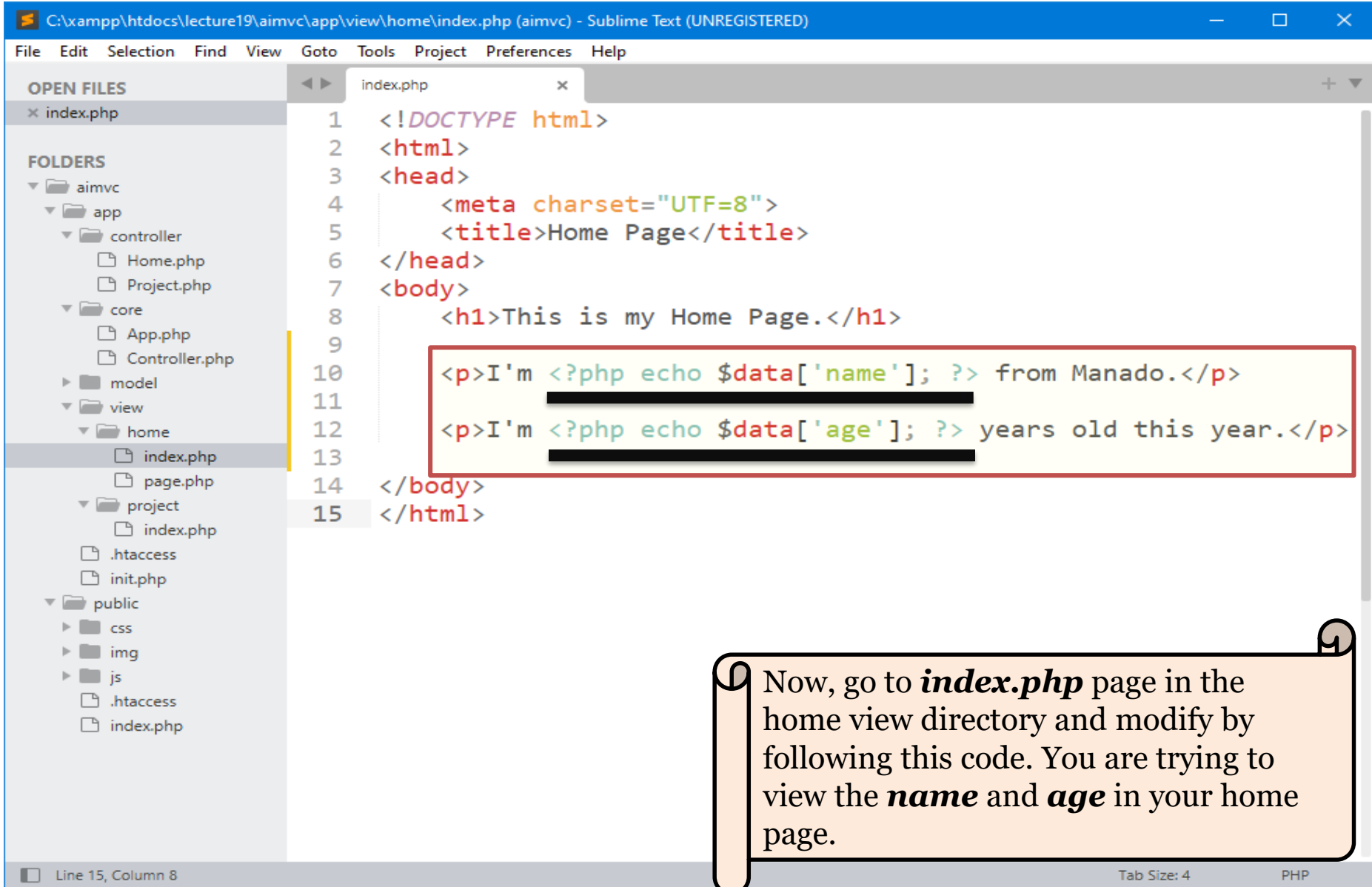
The file explorer on the left shows the project structure. Two arrows point from the file explorer to the code: one from the "home" folder to the "index.php" file, and another from the "project" folder to the "index.php" file.

Now, you can send your data to the target method which can help you to display your data in the target page.

Line 24, Column 6

Tab Size: 4 PHP

3 – Receive data from *display()* Parameter



The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture19\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the following PHP code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
8     <h1>This is my Home Page.</h1>
9
10    <p>I'm <?php echo $data['name']; ?> from Manado.</p>
11
12    <p>I'm <?php echo $data['age']; ?> years old this year.</p>
13
14 </body>
15 </html>
```

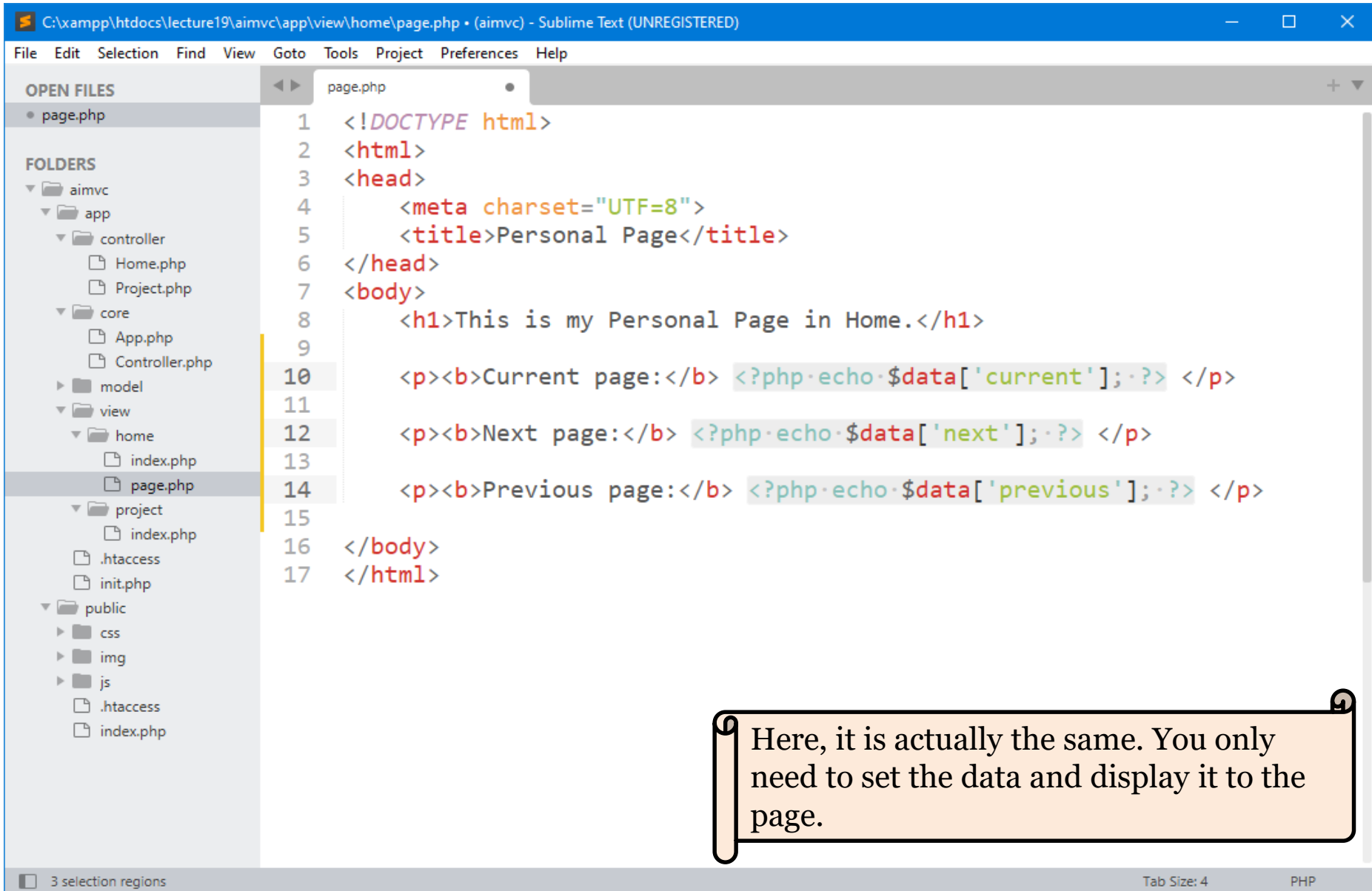
The code on lines 10 and 12 is highlighted with a red box. The code on line 10 is `<p>I'm <?php echo $data['name']; ?> from Manado.</p>` and the code on line 12 is `<p>I'm <?php echo $data['age']; ?> years old this year.</p>`. The code on line 10 is also highlighted with a black box.

The left sidebar shows the "OPEN FILES" and "FOLDERS" panels. The "FOLDERS" panel shows the directory structure: aimvc > app > controller > Home.php, Project.php; core > App.php, Controller.php; model; view > home > index.php, page.php; project > index.php; .htaccess; init.php; public > css, img, js; .htaccess; index.php.

The status bar at the bottom indicates "Line 15, Column 8" and "Tab Size: 4 PHP".

Now, go to **index.php** page in the home view directory and modify by following this code. You are trying to view the **name** and **age** in your home page.

4 – Also Modify *page.php* file in Home Directory



The screenshot shows the Sublime Text editor interface. The title bar indicates the file path: `C:\xampp\htdocs\lecture19\aimvc\app\view\home\page.php`. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The left sidebar displays the 'FOLDERS' tree, showing the project structure with folders like `aimvc`, `app`, `controller`, `core`, `model`, `view`, `home`, `project`, `public`, `css`, `img`, and `js`. The `page.php` file is selected under the `home` folder. The main editor area shows the content of `page.php`, which is an HTML document with PHP code for displaying data.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9
10    <p><b>Current page:</b> <?php·echo·$data['current'];·?> </p>
11
12    <p><b>Next page:</b> <?php·echo·$data['next'];·?> </p>
13
14    <p><b>Previous page:</b> <?php·echo·$data['previous'];·?> </p>
15
16 </body>
17 </html>
```

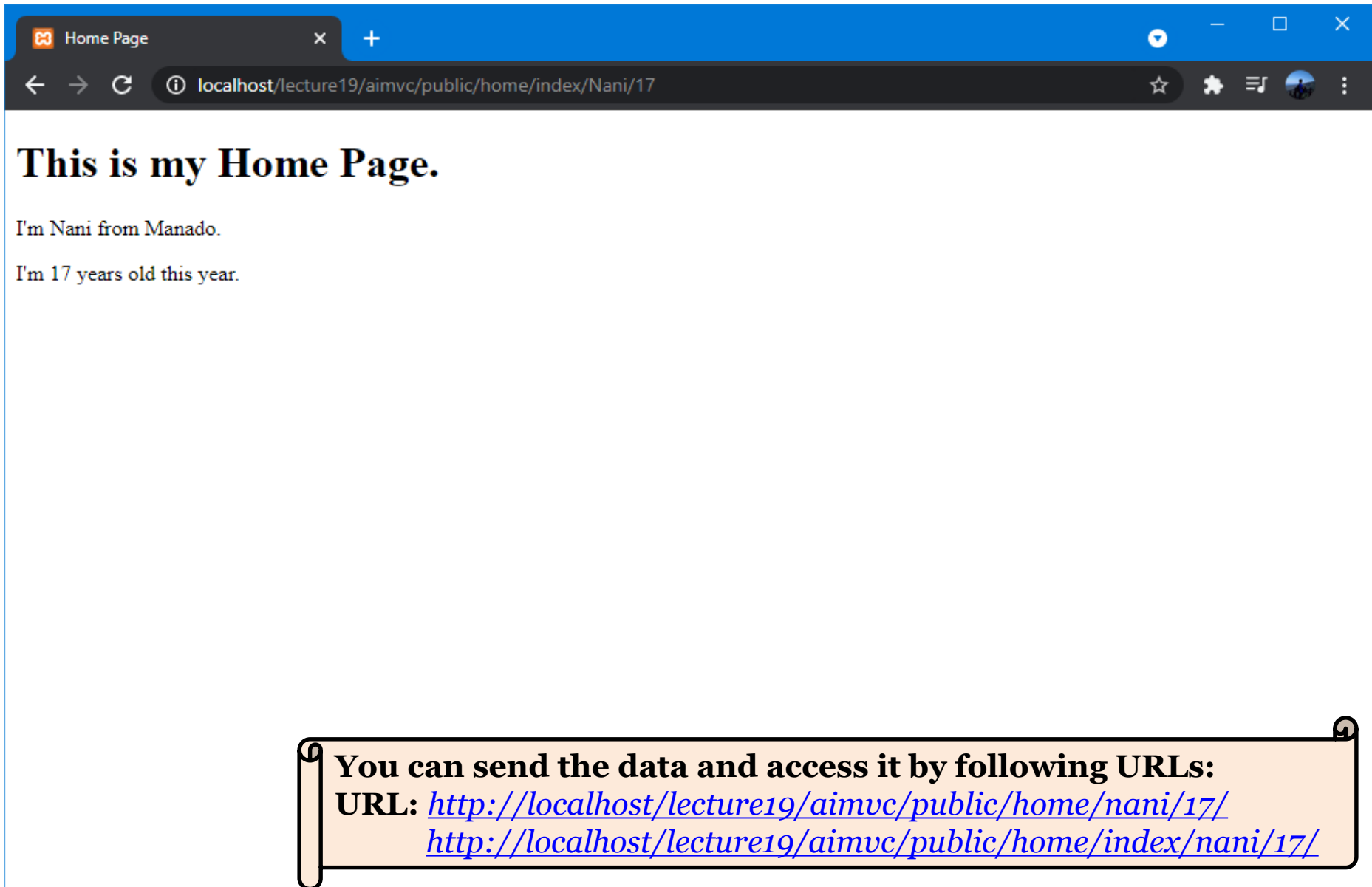
Here, it is actually the same. You only need to set the data and display it to the page.

3 selection regions

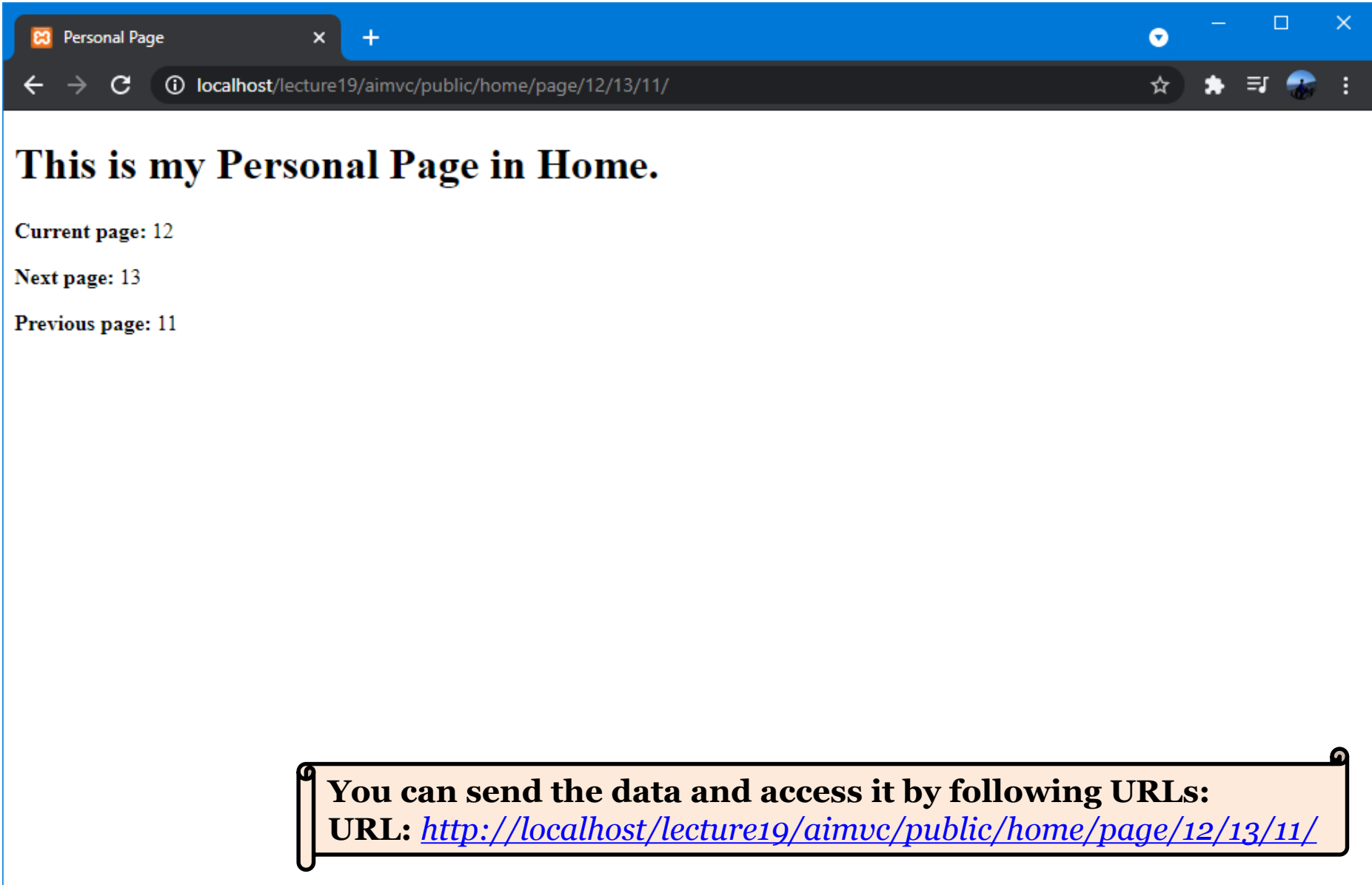
Tab Size: 4

PHP

5 - Testing ⇒ Home Index Page



6 - Testing $\Rightarrow$ Home Personal Page



The screenshot shows a web browser window with a single tab titled "Personal Page". The address bar displays the URL `localhost/lecture19/aimvc/public/home/page/12/13/11/`. The page content includes a heading "This is my Personal Page in Home." followed by three lines of text: "Current page: 12", "Next page: 13", and "Previous page: 11". At the bottom right, a callout box contains a note about sending data and accessing it via URLs, with a specific URL provided.

This is my Personal Page in Home.

Current page: 12

Next page: 13

Previous page: 11

You can send the data and access it by following URLs:
URL: <http://localhost/lecture19/aimvc/public/home/page/12/13/11/>

A decorative border of green leaves and branches at the top of the page.

4 – Create Header & Footer Templates with *their* Title Page

A decorative border of green leaves and branches at the bottom of the page.

HTML Header & Footer Templates

Header



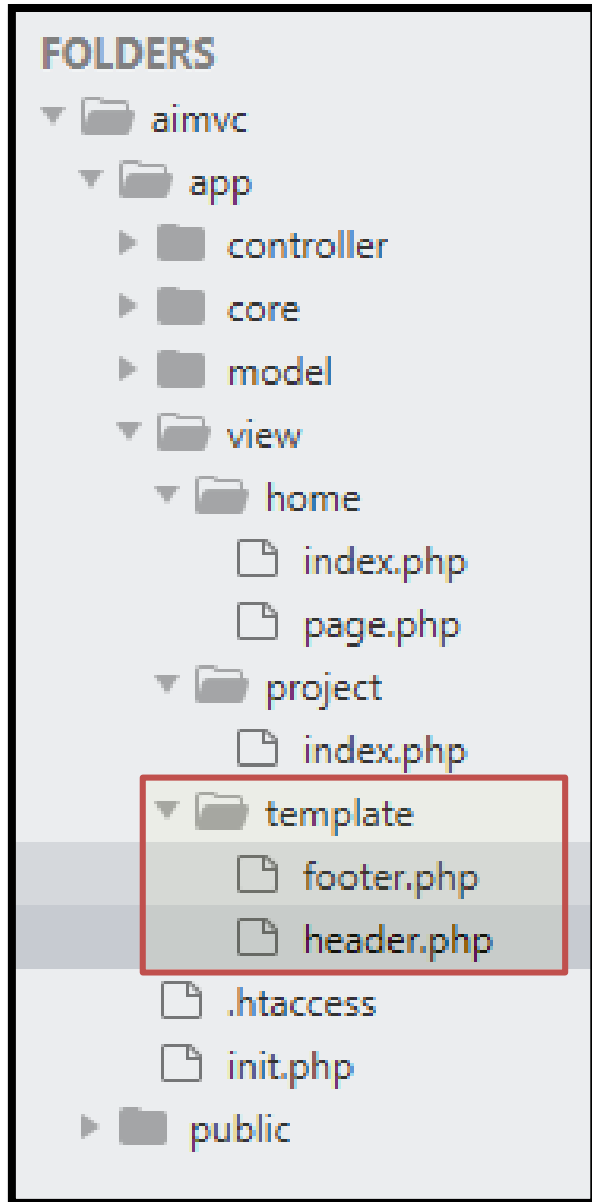
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
8     <h1>This is my Home Page.</h1>
9
10    <p>I'm <?php echo $data['name']; ?> from Manado.</p>
11
12    <p>I'm <?php echo $data['age']; ?> years old this year.</p>
13
14 </body>
15 </html>
```

Footer



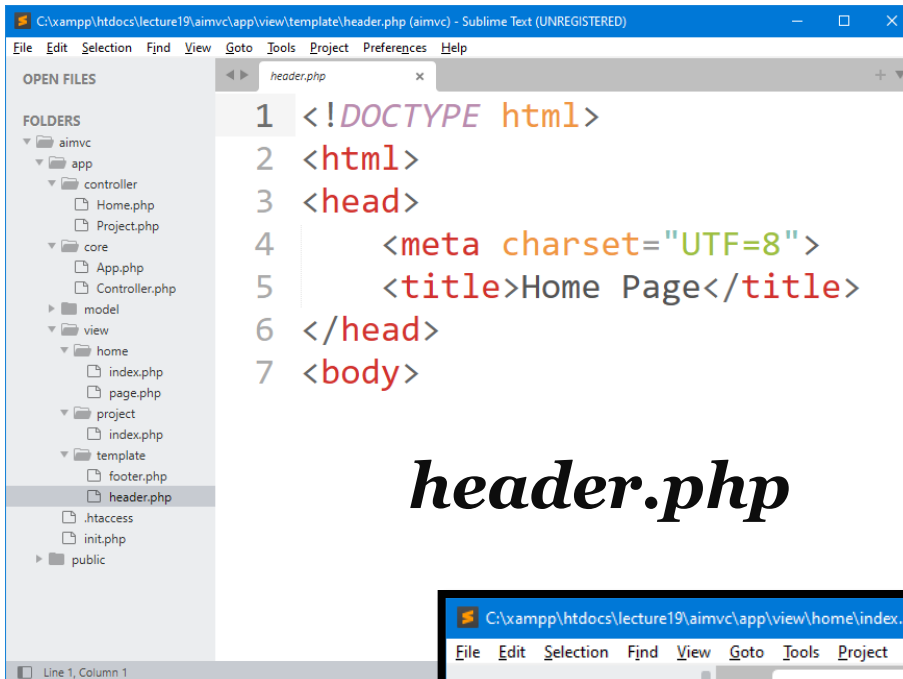
You can easily create sticky or fixed header and footer template using ***the CSS fixed positioning***. Simply apply the CSS position property with the value fixed in combination with the top and bottom property to place the element on the top or bottom of the viewport accordingly.

1 – Create Template Directory in View



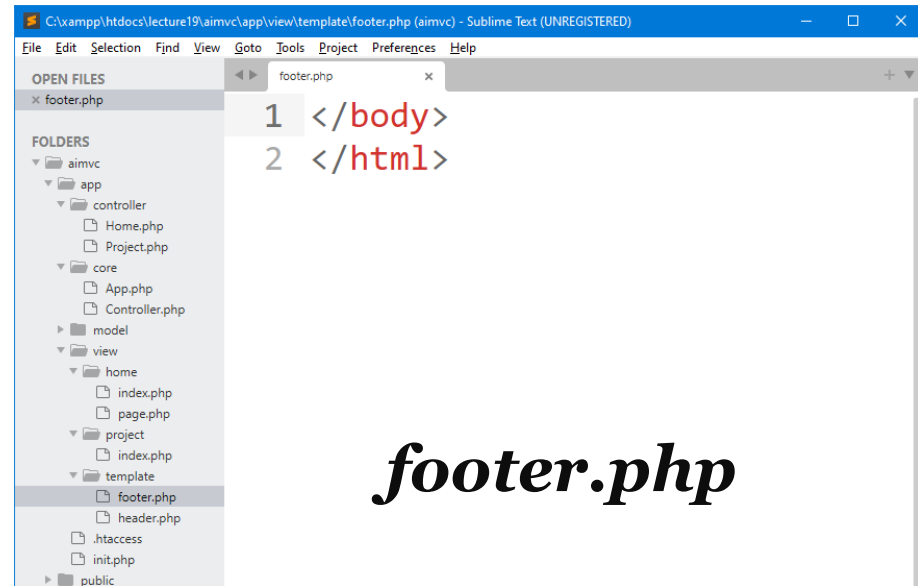
Create one subdirectory in view. This ***template directory*** can store ***header*** and ***footer*** templates and those files can be used as a template for all *php pages* in our framework project.

2 – Header, Footer & Content Sections



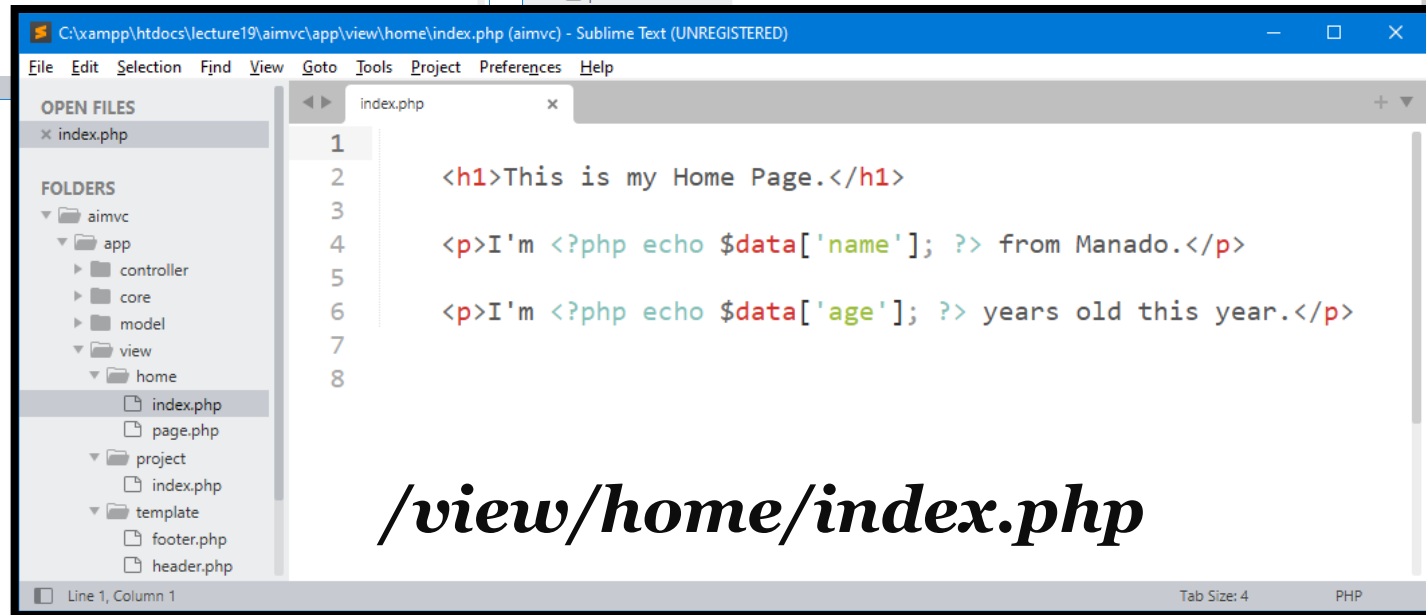
```
C:\xampp\htdocs\lecture19\aimvc\app\view\template\header.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
OPEN FILES
header.php
FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
      App.php
      Controller.php
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
      .htaccess
      init.php
      public
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
```

header.php



```
C:\xampp\htdocs\lecture19\aimvc\app\view\template\footer.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
OPEN FILES
footer.php
FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
      App.php
      Controller.php
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
      .htaccess
      init.php
      public
1 </body>
2 </html>
```

footer.php



```
C:\xampp\htdocs\lecture19\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
OPEN FILES
index.php
FOLDERS
aimvc
  app
    controller
    core
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
1
2 <h1>This is my Home Page.</h1>
3
4 <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5
6 <p>I'm <?php echo $data['age']; ?> years old this year.</p>
7
8
```

/view/home/index.php

3 – Add Header & Footer in Controller

```
C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Home.php

FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
      .htaccess
      init.php
    public

1  <?php
2  class Home extends Controller{
3      // Constructor
4      public function __construct(){
5
6      }
7
8      // Default method
9      public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         // Display page and send data
14         $this->display('template/header');
15         $this->display("home/index", $arr_data);
16         $this->display('template/footer');
17     }
18
19     public function page($current_page=2, $next_page=3, $prev_page=1){
20         // Associative Arrays (arrays with keys)
21         $arr_data['current'] = $current_page;
22         $arr_data['next'] = $next_page;
23         $arr_data['previous'] = $prev_page;
24         // Display page and send data
25         $this->display('template/header');
26         $this->display("home/page", $arr_data);
27         $this->display('template/footer');
28     }
29
30     // Get status
31     private function _status(){
32         echo "This is home/status.";
33     }
34 }
35 ?>
```

Header

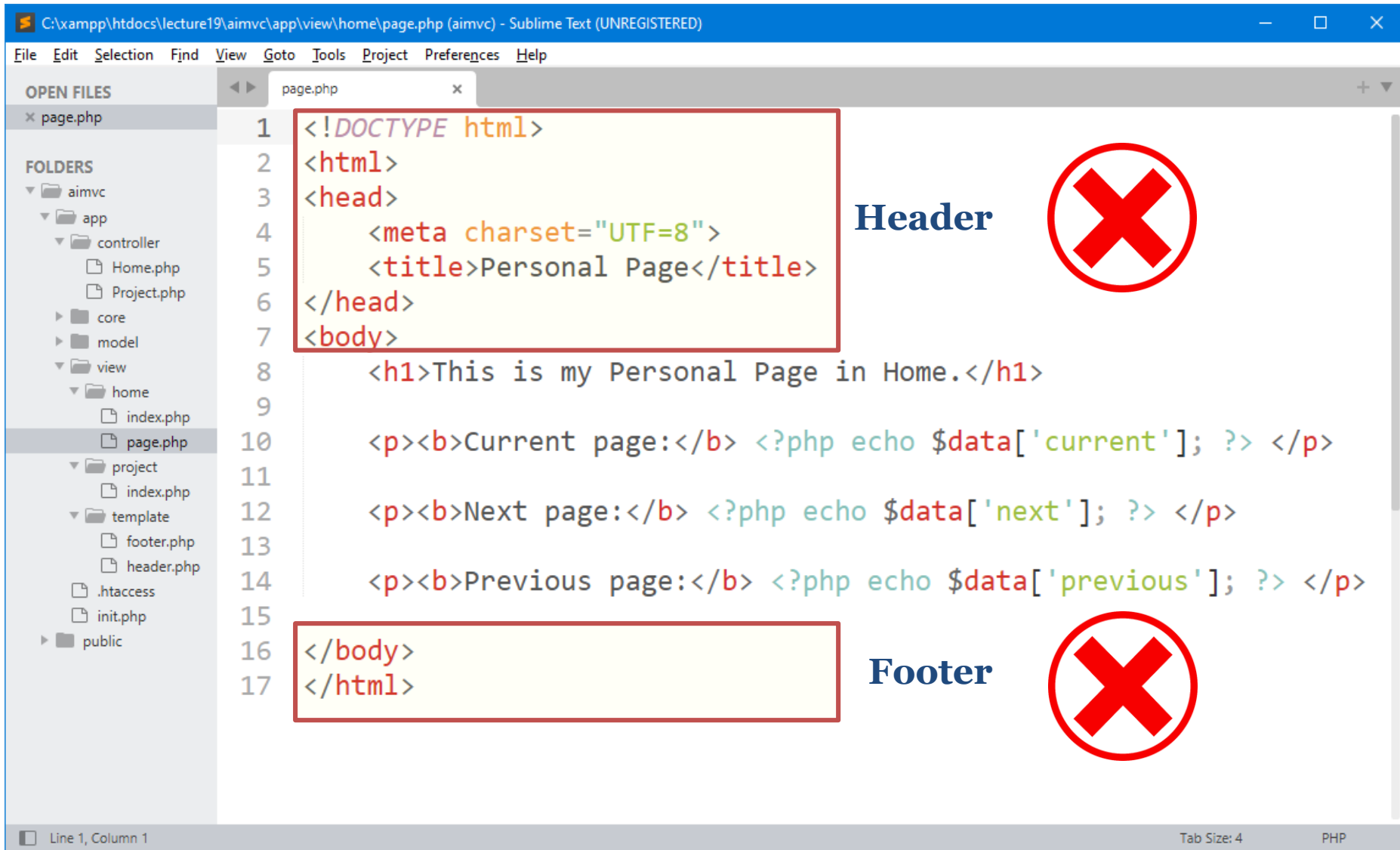
Footer

Header

Footer

4 – Don't Forget !!!

You may need to remove header and footer in **/view/home/page.php** file.



Sublime Text (UNREGISTERED) - C:\xampp\htdocs\lecture19\aimvc\app\view\home\page.php (aimvc)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES

- page.php

FOLDERS

- aimvc
 - app
 - controller
 - Home.php
 - Project.php
 - core
 - model
 - view
 - home
 - index.php
 - page.php
 - project
 - index.php
 - template
 - footer.php
 - header.php
 - .htaccess
 - init.php
 - public

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9
10    <p><b>Current page:</b> <?php echo $data['current']; ?> </p>
11
12    <p><b>Next page:</b> <?php echo $data['next']; ?> </p>
13
14    <p><b>Previous page:</b> <?php echo $data['previous']; ?> </p>
15
16 </body>
17 </html>
```

Header ❌

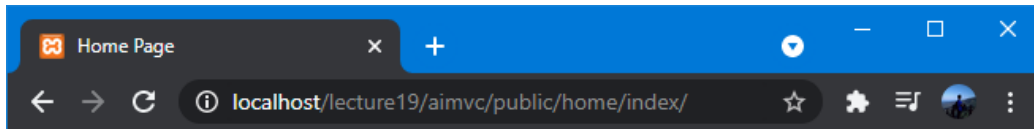
Footer ❌

Line 1, Column 1

Tab Size: 4

PHP

5 - Testing ⇒ Home and Personal Pages

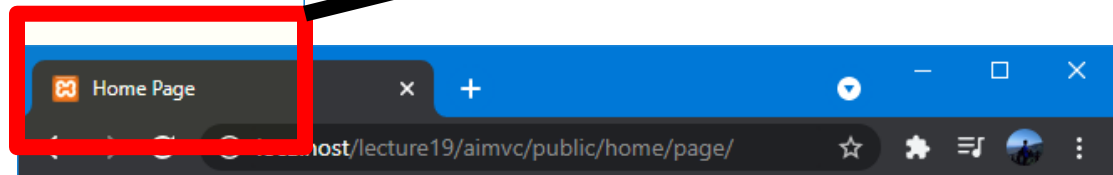


This is my Home Page.

I'm Juan from Manado.

I'm 24 years old this year.

Wrong here !!!



This is my Personal Page in Home.

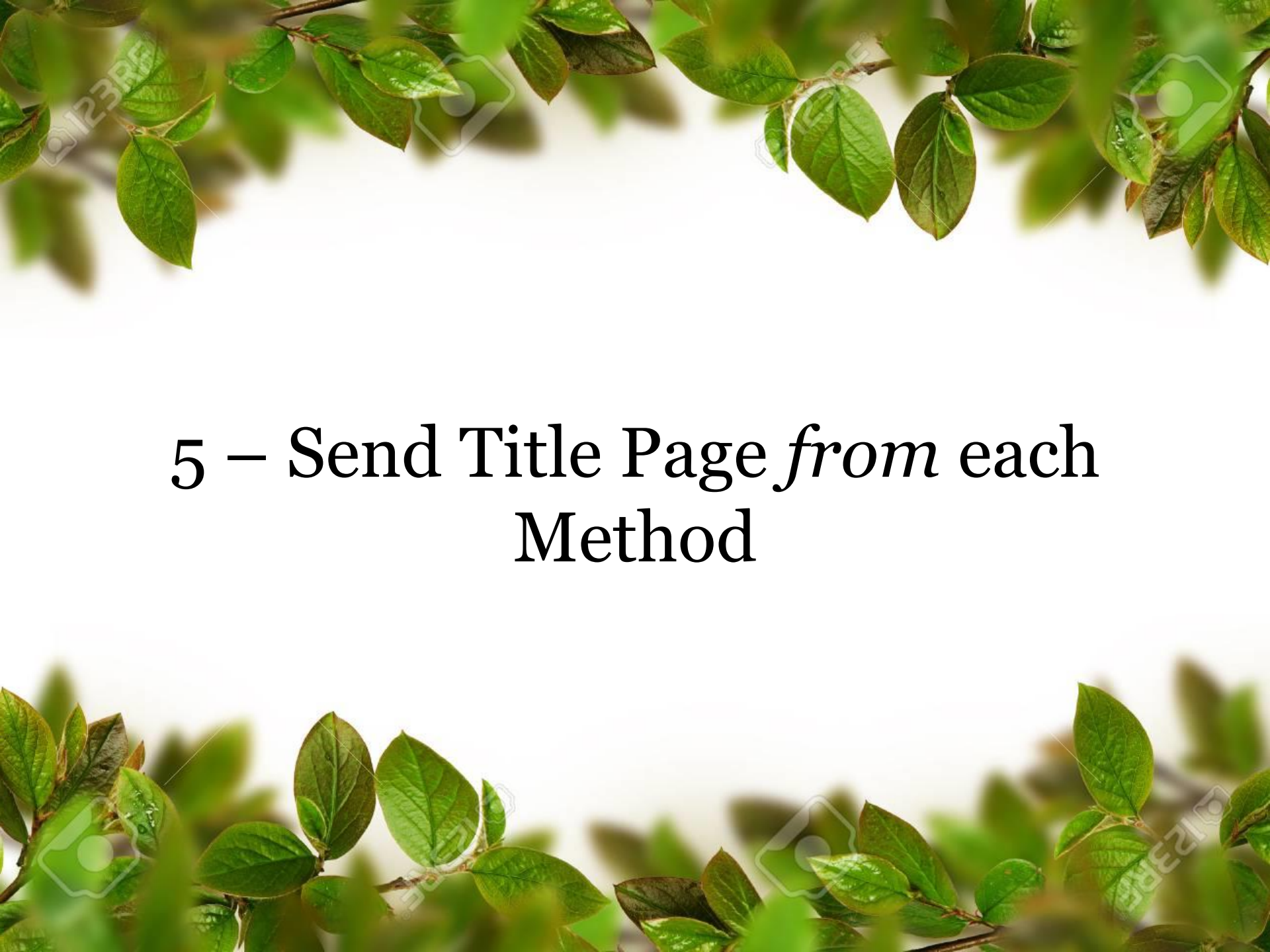
Current page: 2

Next page: 3

Previous page: 1

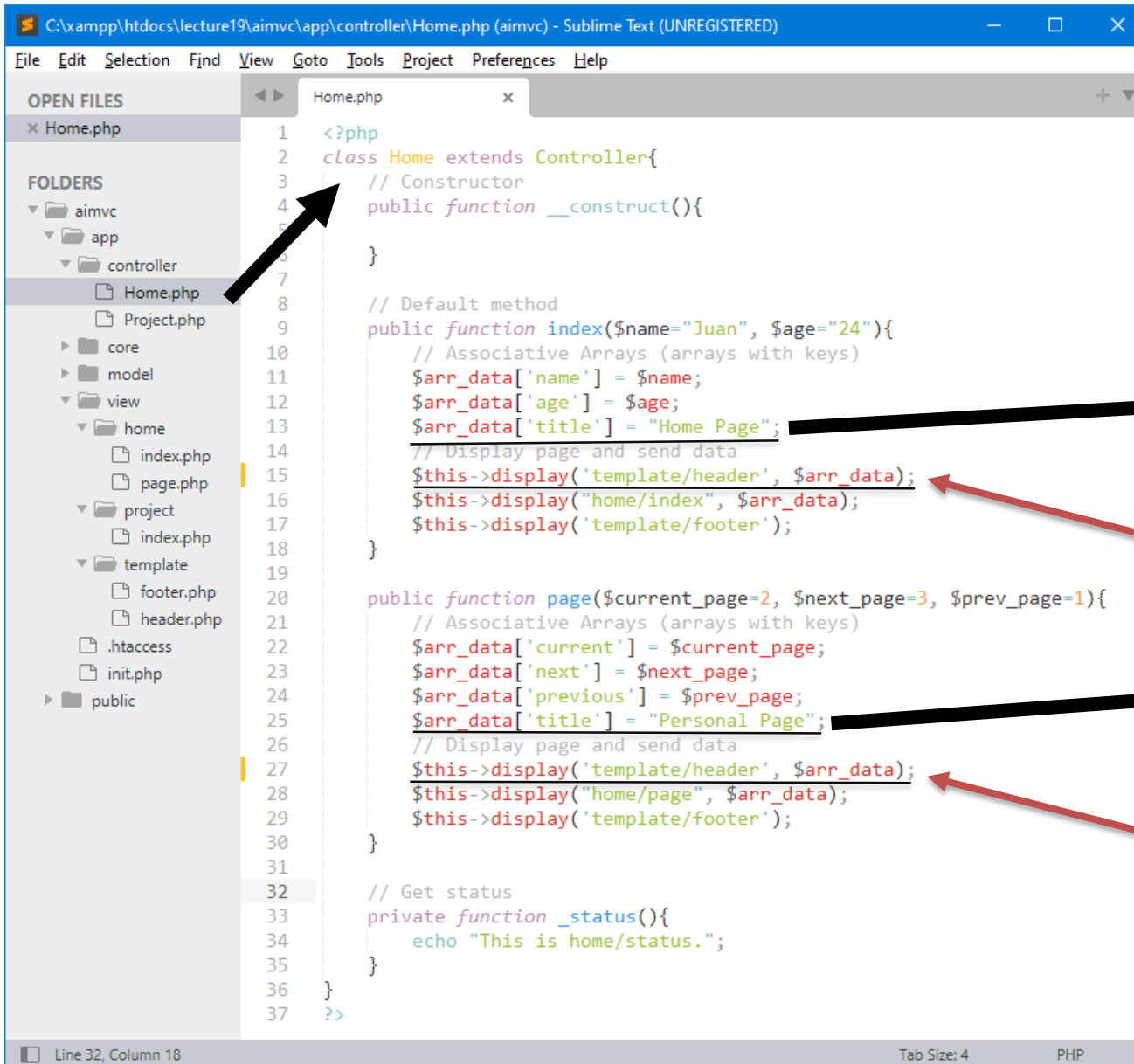
Problem:

Both pages still display the same title page. If you check from previous slide. Our ***/view/home/page.php*** file assigned with title page "Personal Page".

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

5 – Send Title Page *from* each
Method

1 – Send Title Page using Associative Arrays



```
1 <?php
2 class Home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         $arr_data['title'] = "Home Page";
14         // Display page and send data
15         $this->display('template/header', $arr_data);
16         $this->display("home/index", $arr_data);
17         $this->display('template/footer');
18     }
19
20     public function page($current_page=2, $next_page=3, $prev_page=1){
21         // Associative Arrays (arrays with keys)
22         $arr_data['current'] = $current_page;
23         $arr_data['next'] = $next_page;
24         $arr_data['previous'] = $prev_page;
25         $arr_data['title'] = "Personal Page";
26         // Display page and send data
27         $this->display('template/header', $arr_data);
28         $this->display("home/page", $arr_data);
29         $this->display('template/footer');
30     }
31
32     // Get status
33     private function _status(){
34         echo "This is home/status.";
35     }
36 }
37 ?>
```

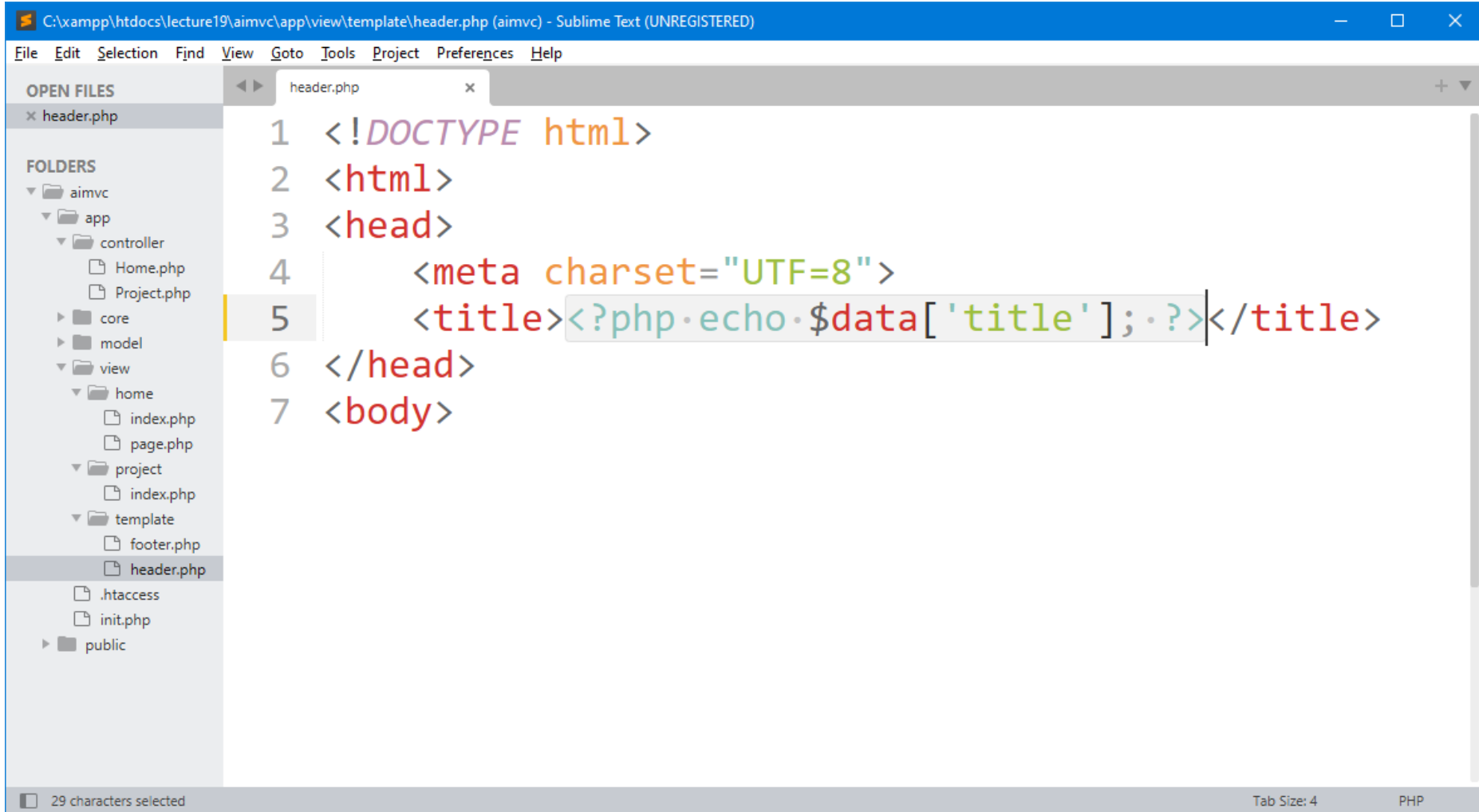
Title Page

Array data also need to send as a parameter in **display function**.

Title Page

Array data also need to send as a parameter in **display function**.

2 – Modify *header.php* file in Title Tag



```
C:\xampp\htdocs\lecture19\aimvc\app\view\template\header.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
header.php
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title><?php·echo·$data['title'];·?></title>
6 </head>
7 <body>
```

29 characters selected

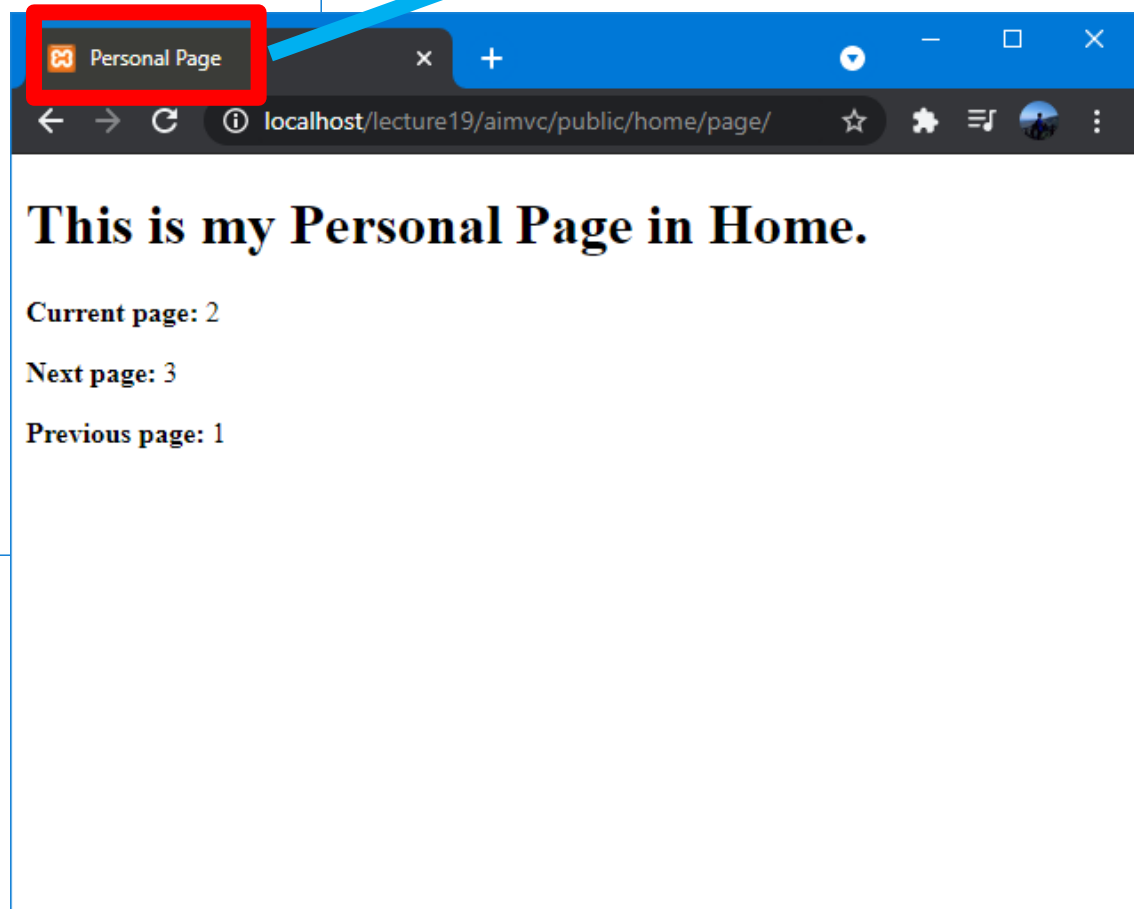
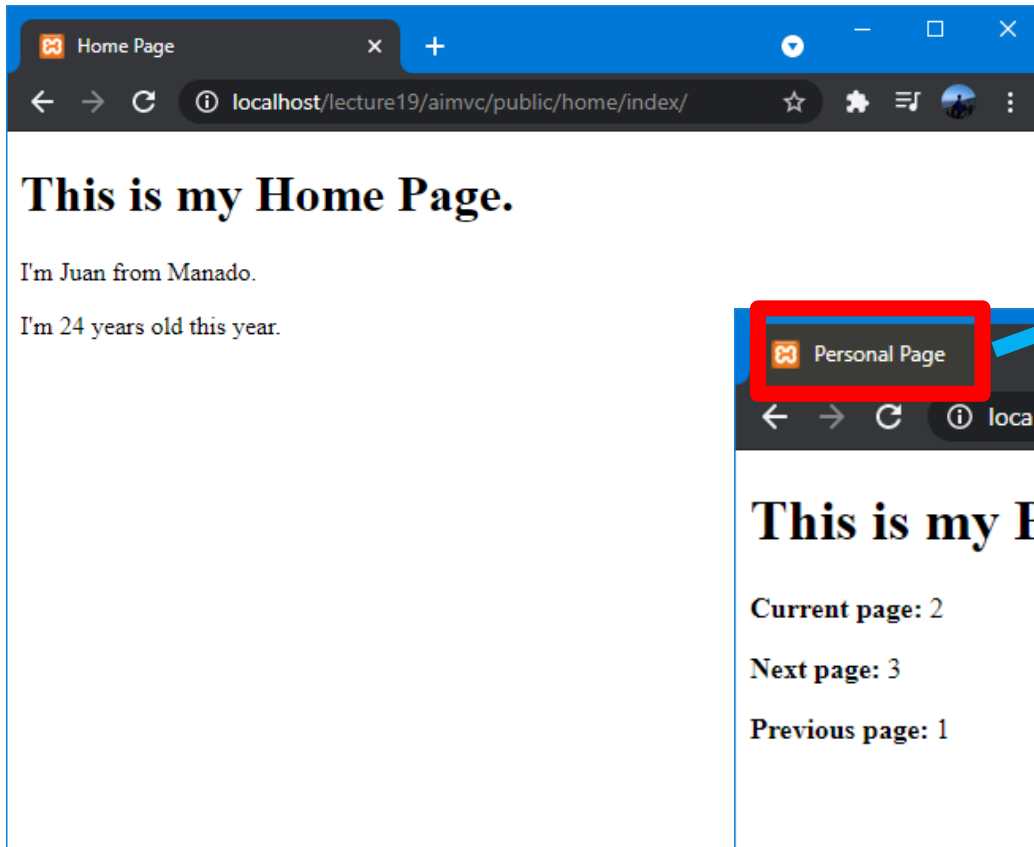
Tab Size: 4 PHP

Change following line of code:

From : `<title>Home Page</title>`

To : `<title><?php echo $data['title']; ?></title>`

3 - Testing ⇒ Home and Personal Pages



It's correct Now !!!



exercise for students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture19\
Create file : `_.php`

Buat semua contoh yang sudah di pelajari sebelumnya di *lecture* ini.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

